MARIANO FRANCISCO RODRIGUEZ

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OBJECTIVE

Looking for opportunities to learn how to build high performing, scalable, maintainable, and idiomatic solutions to software industry problems. Passionate about learning the methodologies that comprise the foundations of successful engineering teams, specifically in the context of embedded systems teams.

EDUCATION

- **University of Illinois at Urbana-Champaign** *Expected December 2027*
  BS, Computer Science, Grainger College of Engineering Urbana, Illinois
  - **GPA**: 3.93/4.00
  - **Relevant Coursework**: Discrete Math, Data Structures, Computer Architecture, Numerical Methods

SKILLS

- **Programming Languages:** Rust, C++, Python, JavaScript, C, OCaml, Java, Kotlin, SQL
- **Web Technologies:** NextJS, React, Django, Flask, REST, WebSockets, HTML, CSS
- **DevOps:** Git, GitHub Actions, Cargo, Bash, Make, Docker
- **Other Technologies:** Linux, UART, PWM, I2C, SPI, Verilog, PostgreSQL, SQLite

EXPERIENCE

- **Operation Autopilot, via Discovery Partners Institute** *June 2025 - August 2025*
  Summmer Tech Intern Chicago, Illinois
  - Wrote Python-based leads generation software for prospecting potential customers, generating over 100 new leads and leading to more potential revenue
  - Developed Flask interface for leads generator, using webcrawling, and Google Places API to enrich data
- **Mundelein High School** *June 2023 - July 2023*
  Camp Counselor Mundelein, Illinois
  - Developed weekly lessons teaching middle schoolers the Python programming language
  - Added interactivity to learning via weekly projects and discussions around coding logic
- **Discovery Partners Institute** *June 2022 - August 2022*
  Intern Chicago, Illinois
  - Developed mobile app end user app as part of capstone project, honing own skills with Swift
  - Implemented SQLite database to store user app data

PROJECTS

- **Outdoorzy: Social App for Finding Workout/Sports Partners/Groups** *June 2025 - Present*
  Python, Django, REST [LinkedIn]
  - Helped design system architecture for Geolocation-based CRUD app
  - Co-Developing backend CRUD REST API along with token-based authentication layer
- **Illinois Space Society: Solid Fuel Rocket Propulsion Testing Rig Software** *August 2024 - Present*
  Rust, SPI, Linux, Device Tree, Git [Blog Post]
  - Developed Rust binary for Raspberry Pi Zero 2 to collect time-series data from SPI sensors at 1 kHz
  - Wrote bespoke device tree for Raspberry Pi Real-Time Linux kernel to reconfigure hardware GPIO pins into SPI pins
  - Oversaw live rocket motor hotfire, with 100% uptime for telemetry
  - Currently overseeing LoRa integration in cross-functional team with intention of achieving a kilometer safe zone
- **parse_rc_ibus: Parser for IBUS radio protocol** *August 2024 - November 2024*
  Rust, UART [Crates.io]
  - Designed robust data model, ensuring checksum validity and developer ergonomics via 100% test coverage
  - Published library on crates.io, receiving over 3000 downloads
- **VexDoc: HTML Documentation Generating Command-line Interface** *June 2024 - July 2024*
  Rust [GitHub]
  - Designed Unix style command-line interface to generate HTML code documentation
  - Used a finite-state machine to parse annotations in the target source code
- **Chatter: AOL Instant Messenger Inspired WebSocket Chat App** *September 2023 - December 2023*
  Rust, WebSockets, HTML, CSS [GitHub]
  - Developed WebSocket chat app using Rust as server
  - Coordinated multiple development efforts between classmates who were interested in contributing

INVOLVEMENTS & MEMBERSHIPS

- Illinois Space Society, Society of Hispanic Professional Engineers, Peruvian Student Association, Association for Computing Machinery, National Marrow Donor Program